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ID publico: case_0009

Paginas del documento: 4

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provided with a single brand enzymatic cleaner (Beaver *et al.*, 1989; Melese, 1994). On the first day all areas of house with urine odor had to be cleaned with the mentioned cleaner. In addition, other litter boxes and food containers were removed to remain just one per cat. Beside that they were asked to clean the litter box from the feces and urine daily and it is better if they clear the whole litter box every week.

The placing of litter box should have been same and they should refrain from physically or verbally punishing the cat for urine marking. For the achievement of more assurance, we telephoned every day for monitoring and asking about the procedure. After this phase that we named Phase 1, the selected cats were cleaved in two-way blind trial, with half of cats receiving placebo and half receiving medication, and a description of the drug being tested were explained. The cats were randomly selected to be in the drug or placebo group with an investigator who had no direct contact with the cat or owners.

The dosage of venlafaxine was considering one mg per kg, PO, q24hrs (Dehasse, 1997) and the placebo group took sterile water as a drop for 3 droplets 3 times per day. To prevent unwanted reactions in drug group we started with 0.5mg per kg per day for 4 days and we did not calculate these 4 days as treatment duration. We consider the first treatment phase two weeks according to most psychiatric medication for deciding if the drug is effective or not. After that if any achievement was shown, we continued the treatment for 8 weeks and sometimes more depending on resolving the problem. If any achievement was not seen during the treatment period, we tapered the usage of drug by advising the 0.5mg per kg per day intake for 4 days. Only the samples that showed more than 70 percent improvement were advised to continue the treatment.

The mean weekly number of marks during the 2 week baseline period was used as a reference with which to compare the marking rate of cats throughout 2 weeks of treatment. So, each cat served as its own control. A t-test was used to judge the overall effects of treatment from baseline. Statistical procedure followed standard established methodology¹⁷ and a model or variable was considered statistically significant if $P \leq 0.05$.

The cats entered the study from three clinics in Tehran, IRAN. Since the behavior Urine Marking considered to be normal in male cats, only the male cats that have been castrated and lived indoor, were selected to entered the study. All the samples were evaluated by ultrasonography and CBC (appendix1-2). Studied Cats had to have at least 6 instances of vertical urine marking per week while not receiving any medication for it to be a behavioral problem. Vertical Urine marking was recorded by owners on a daily basis for two weeks by date. To remove urine odor all the owners were

RESULTS

At the goal enrollment, 20 cats that were 1 year old and 10 per week were entered in the study and they were separated into two groups of treatment and placebo group. For easy evaluation of results, we numbered the cats from 1 to 20 where the 1 to 10 were placed on control group and the 11 to 20 stayed in the treatment group. Four of the samples in CBC presented leukocytosis that according to absence of clinical signs, no usage of glucocorticoids and the result of ultrasonography and also conformity of them to stress leukogram were considered related to stress on sampling time. In addition, the results of complete urinalysis and ultrasonography evaluation all showed the normal condition of samples.

In the placebo group, there was a mean ($\pm$ SE) of 10.2 ($\pm$ 0.66) marks per week, and the treated group had a mean of 11.2 ($\pm$ 0.64) marks per week (Table 1). There was a significant decrease in marking episodes with both groups when the rate during the treatment phase was compared with that during the baseline. Almost all the members of treatment group showed more degrees of decreasing urine marking in the 2nd week than we imagined ($P < 0.0001$). The placebo group mean reached the number of 7.8 ($\pm$ 0.81) and the treatment one was 2.9 ($\pm$ 0.58) (Table 1). No one in the treated group showed achievement under 70 percent and so we continued the treatment with the same dose for another 2 weeks. There were cats that showed 1 or 2 urine marking the 2nd week of treatment.

Table 1. Evaluation of body weight change and urine marking before and after of study

Experimental group	Urine marking	Body weight change
<i>Placebo group</i>		
Before	10.20 $\pm$ 0.66	3.55 $\pm$ 0.37
After	7.80 $\pm$ 0.81	3.53 $\pm$ 0.37
P Value	0.021	0.54
<i>Treatment group</i>		
Before	11.20 $\pm$ 0.64	4.36 $\pm$ 0.31
After	2.90 $\pm$ 0.58	4.22 $\pm$ 0.29
P value	<0.0001	0.100

DISCUSSION

The results between the response of cats receiving venlafaxine and placebo was strongly different and our results made the difference clear. The lack of response of the placebo group was significant in comparison with treatment group. The decrease in placebo group mean urine marking might be related to management and standardized environmental factors. The psychiatric literature showed that the SSNRI may require a minimum of two weeks or more to judge the result (McEvoy, 1993; SM, 2000). The response of cats to venlafaxine in this study showed more achievement observed in open label studies on diazepam (Cooper and Hart, 1992; Marder, 1991) and buspirone (Hart *et al.*, 1993). That may be related to the number of samples and the single gender that we selected.

In one study, the researchers published the effects of clomipramine on cats presented for Urine Marking. In 25 cats the effects were

statistically significant but the effects were seen after 4 weeks (Landsberg and Wilson, 2005). The effects of using pheromones in different studies suggest that it may be as effective or more (Frank *et al.*, 1999), but they are not reachable in some countries like middle east or they may be so expensive for owners beside side effects like crystalluria or involvement of the urogenital system.

Associated with other parameters evaluated in this study like changes in weight, appetite, sleep pattern, balance and physical treatment and alimentary tract problems, blood biochemistry profile and CBC were retested to establish if any side effect exists or not. Generally, in both groups, changes in weight were not seen, but in 4 cases of treatment group there was a little decrease in appetite. In 3 cases of treatment group, there was a change in sleep pattern and seemed sleep more than before. One of them had a tremor, and diarrhea was seen in 2 cases beside the constipation in 3 cases and 3 of treatment

ncrease in hepatic enzymes that they reject to the normal value.

know about the environmental problems that may lead or worsen the condition, and medical therapy should be performed after standardizing the factors previously mentioned. For those cats that continue urine marking after environmental management, treatment with venlafaxine (one mg per kg per PO, q24 hrs.) can reduce or solve the problem. The positive results can be seen during the first 2 weeks and duration of treatment is case-dependent, guided by the principle that most cats should be treated more than 8 weeks with this important notice that in any condition, tapering of drug intake is very important in psychiatric therapy.

REFERENCES

- BEAVER, B.V. Housesoiling by cats: a retrospective study of 120 cases. *J. Am. Anim. Hosp. Assoc.*, v.25, p.631-637, 1989.
- BEAVER, B.V.; TERRY, M.L.; LASAGNA, C.L. Effectiveness of products in eliminating cat urine odors from carpet. *J. Am. Vet. Med. Assoc.*, v.194, p.1589-1591, 1989.
- BORCHELT, P.L. Cat elimination behavior problems. *Vet. Clin. North Am. Small Anim. Pract.*, v.21, p.257-264, 1991.
- COOPER, L.; HART, B.L. Comparison of diazepam with progestin for effectiveness in suppression of urine spraying behavior in cats. *J. Am. Vet. Med. Assoc.*, v.200, p.797, 1992.
- DEHASSE, J. Feline urine spraying. *Appl. Anim. Behav. Sci.*, v.52, p.365-371, 1997.
- ECKSTEIN, R.A.; HART, B.L. Pharmacologic approaches to urinemarking in cats. *Psychopharmacol. Anim. Behav. Disord. Malden Mass Blackwell Sci.*, v.26, p.264-276, 1998.
- FRANK, D.F.; ERB, H.N.; HOUP, K.A. Urine spraying in cats: presence of concurrent disease and effects of a pheromone treatment. *Appl. Anim. Behav. Sci.*, v.61, p.263-272, 1999.
- HART, B.L. Behavioral and pharmacologic approaches to problem urination in cats. *Vet. Clin. Small Anim. Pract.*, v.26, p.651-658, 1996a.
- HART, B.L. Interpreting published results of extra-label drug use with special reference to reports of drugs used to correct problem behavior in animals. *J. Am. Vet. Med. Assoc.*, v.209, p.1382-1385, 1996b.
- HART, B.L.; COOPER, L. Factors relating to urine spraying and fighting in prepubertally gonadectomized cats. *J. Am. Vet. Med. Assoc.*, v.184, p.1255-1258, 1984.
- HART, B.L.; ECKSTEIN, R.A.; POWELL, K.L.; DODMAN, N.H. Effectiveness of buspirone on urine spraying and inappropriate urination in cats. *J. Am. Vet. Med. Assoc.*, v.203, p.254-258, 1993.
- LANDSBERG, G.M.; WILSON, A.L. Effects of clomipramine on cats presented for urine marking. *J. Am. Anim. Hosp. Assoc.*, v.41, p.3-11, 2005.
- MARDER, A.R. Psychotropic drugs and behavioral therapy. *Vet. Clin. North Am. Small Anim. Pract.*, v.21, p.329-342, 1991.
- MCEVOY, G.K. *American hospital formulary service-drug information 92*. Bethesda, MD: American Society of Hospital Pharmacists, 1993.
- MELESE, P. Detecting and neutralizing odor sources in dog and cat elimination problems. *Appl. Anim. Behav. Sci.*, v.39, p.188-189, 1994.
- METZ, D.; MEDAM, T.; MASSON, S. Double-blind, placebo-controlled trial of venlafaxine to treat behavioural disorders in cats: a pilot study. *J. Feline Med. Surg.*, v.24, p.539-549, 2022.
- OLM, D.D.; HOUP, K.A. Feline house-soiling problems. *Appl. Anim. Behav. Sci.*, v.20, p.335-345, 1988.
- PRYOR, P.A.; HART, B.L.; CLIFF, K.D.; BAIN, M.J. Effects of a selective serotonin reuptake inhibitor on urine spraying behavior in cats. *J. Am. Vet. Med. Assoc.*, v.219, p.1557-1561, 2001.
- PRYOR, P.A.; HART, B.L.; BAIN, M.J.; CLIFF, K.D. Causes of urine marking in cats and effects of environmental management on frequency of marking. *J. Am. Vet. Med. Assoc.*, v.219, p.1709-1713, 2001.
- S Mc Cune-The UFAW handbook on the Care and Management of Laboratory and Other Research Animals. 8th ed. 2010, pp453-472.

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